SOC MONITOR LAB
WAZUH SIEM PROJECT REPORT

Ubuntu Server • Kali Linux Agent • Custom Alert Detection

Project repository: https://github.com/AKASH456-dev/soc-monitor-lab

1. Project Overview

SOC Monitor Lab is a small home-lab project built to demonstrate centralized security monitoring with Wazuh SIEM. An Ubuntu Server acts as the Wazuh manager/indexer/dashboard host, while a Kali Linux virtual machine runs the Wazuh agent. The lab collects endpoint events and verifies a custom alert rule.

Project repository: https://github.com/AKASH456-dev/soc-monitor-lab

2. Objectives

- Deploy Wazuh SIEM components on Ubuntu Server.
- Connect a Kali Linux endpoint as a Wazuh agent.
- Generate a test event from Kali Linux and observe it in Threat Hunting.
- Create and validate a custom Wazuh rule with ID 100100.
- Record screenshots and commands as project evidence.

3. Lab Environment

Component	Configuration
Wazuh server	Ubuntu Server 22.04.5 LTS
SIEM version	Wazuh 4.14.x (README records Wazuh 4.14)
Dashboard URL	https://192.168.31.44 (works on the same reachable network)
Agent OS	Kali Linux
Agent name	kali-agent
Agent ID	001
Custom rule ID	100100
Custom alert level	5

Custom alert text	SOC Lab: Custom test event received from agent

Note: IP addresses can change when the virtual machines connect to a different router/network. If the dashboard URL stops working, check the current Ubuntu server IP and use that address.

4. Architecture and Event Flow

Kali Linux → Wazuh agent → Ubuntu Wazuh manager → Wazuh indexer → Wazuh Dashboard / Threat Hunting.

The agent forwards endpoint logs to the manager. Wazuh decodes and evaluates events against rules. Matching events are written to alerts.json and indexed so they can be searched in the dashboard.

5. Installation and Setup Commands

5.1 Ubuntu Server — basic checks

```
lsb_release -a
df -h
ip addr
sudo systemctl status wazuh-indexer --no-pager
sudo systemctl status wazuh-manager --no-pager
sudo systemctl status wazuh-dashboard --no-pager
```

Purpose: confirm the Ubuntu release, available disk space, server IP address, and service status. In this lab, disk space was an issue during troubleshooting, so checking df -h is important.

5.2 Download the Wazuh installation assistant

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
sudo bash ./wazuh-install.sh -a
```

Run only on the intended Ubuntu Wazuh server and follow the official Wazuh installation instructions for the selected release. The -a option performs the all-in-one installation workflow. If the system already has components installed, do not blindly rerun the installer; inspect service status and the official upgrade/reinstallation guidance first.

5.3 Kali Linux — agent status and connectivity

```
sudo systemctl status wazuh-agent --no-pager
sudo systemctl is-active wazuh-agent
sudo /var/ossec/bin/agent_control -lc
```

Expected evidence from this lab: agent_control listed ID 001, name kali-agent, status Active. The command should be run on the Ubuntu Wazuh manager for the manager's connected-agent list.

5.4 Verify sudo events

```
sudo grep -F 'Successful sudo to ROOT executed' /var/ossec/logs/alerts/alerts.json | tail -n
3
```

This query showed Wazuh rule 5402 events, including sudo activity from kali-agent. These events demonstrate that the agent was forwarding endpoint security events.

6. Custom Alert Rule

6.1 Rule configuration

File: /var/ossec/etc/rules/local_rules.xml on the Ubuntu Wazuh manager.

```
<rule id="100100" level="5">
  <match>SOC_LAB_TEST_EVENT</match>
  <description>SOC Lab: Custom test event received from agent</description>
  <group>soc_lab,custom_test,</group>
</rule>
```

The custom rule matches the marker SOC_LAB_TEST_EVENT and creates a level-5 alert. The rule belongs inside the existing rules group in local_rules.xml; do not add a second outer group wrapper.

6.2 Validate and restart the manager after editing

```
sudo grep -n -A 12 -B 5 '100100' /var/ossec/etc/rules/local_rules.xml
sudo systemctl restart wazuh-manager
sudo systemctl is-active wazuh-manager
```

The grep command confirms the rule text and ID. Restart the manager after a rule edit so the rule is loaded. If restart fails, inspect the manager logs before proceeding.

6.3 Generate the test event on Kali Linux

```
logger "SOC_LAB_TEST_EVENT: Kali agent test successful"
```

This sends a test message to the system log. The Wazuh agent collects the event and forwards it to the manager.

6.4 Verify the custom alert on Ubuntu

```
sudo grep -F 'SOC_LAB_TEST_EVENT' /var/ossec/logs/alerts/alerts.json | tail -n 3
```

Verified result in this lab: rule ID 100100, level 5, description 'SOC Lab: Custom test event received from agent', agent kali-agent (ID 001), and full_log containing 'SOC_LAB_TEST_EVENT: Kali agent test successful'.

7. Dashboard Verification

- Open the Wazuh Dashboard at https://192.168.31.44 (or the current server IP).
- Open Threat Hunting → Events.

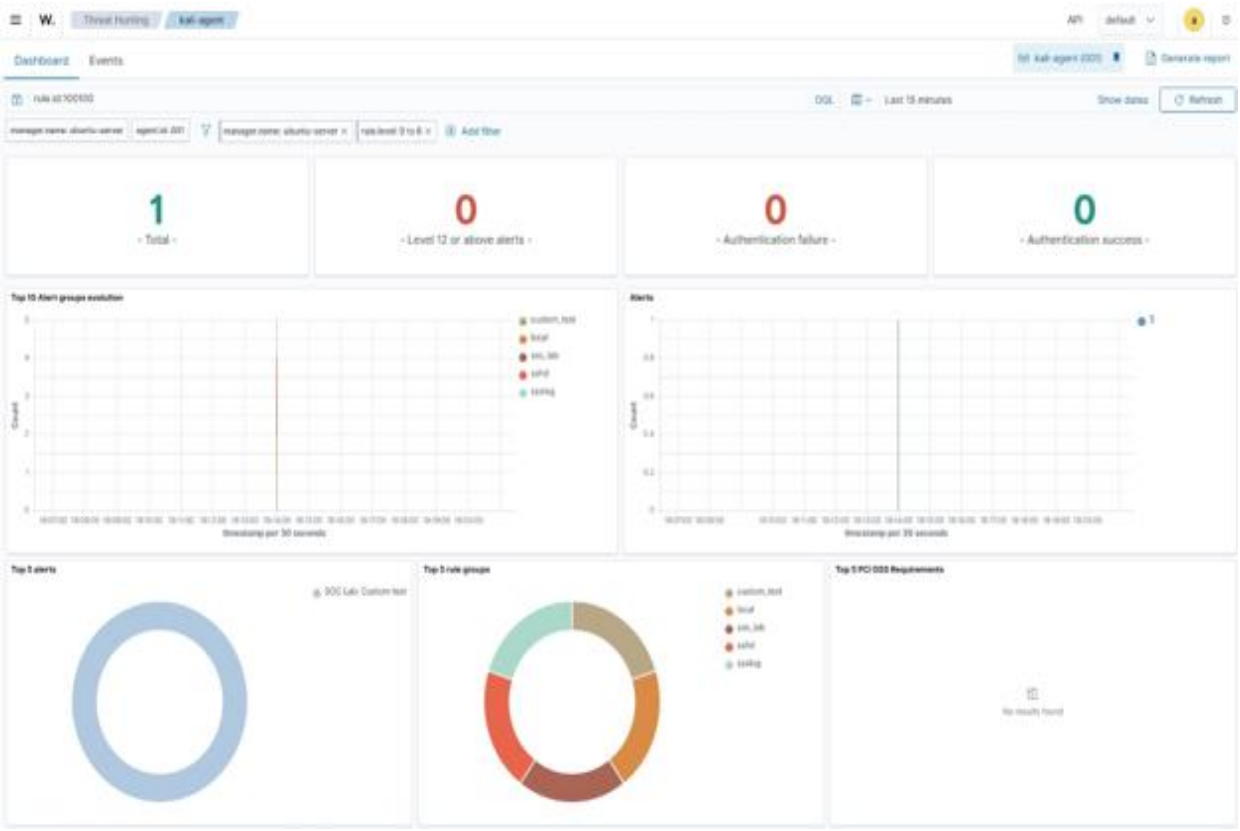
- Search using rule.id:100100 and set the time picker to a period that includes the test event. The event shown in the captured screenshot was at Oct 27, 2026 around 19:14 local dashboard time.
- Confirm the event row shows agent kali-agent, description 'SOC Lab: Custom test event received from agent', rule level 5, and rule ID 100100.

The Events screenshot in this report shows one matching event. The Dashboard view filtered for the Kali agent shows one alert and the custom_test / soc_lab groups.

8. Screenshots and Evidence

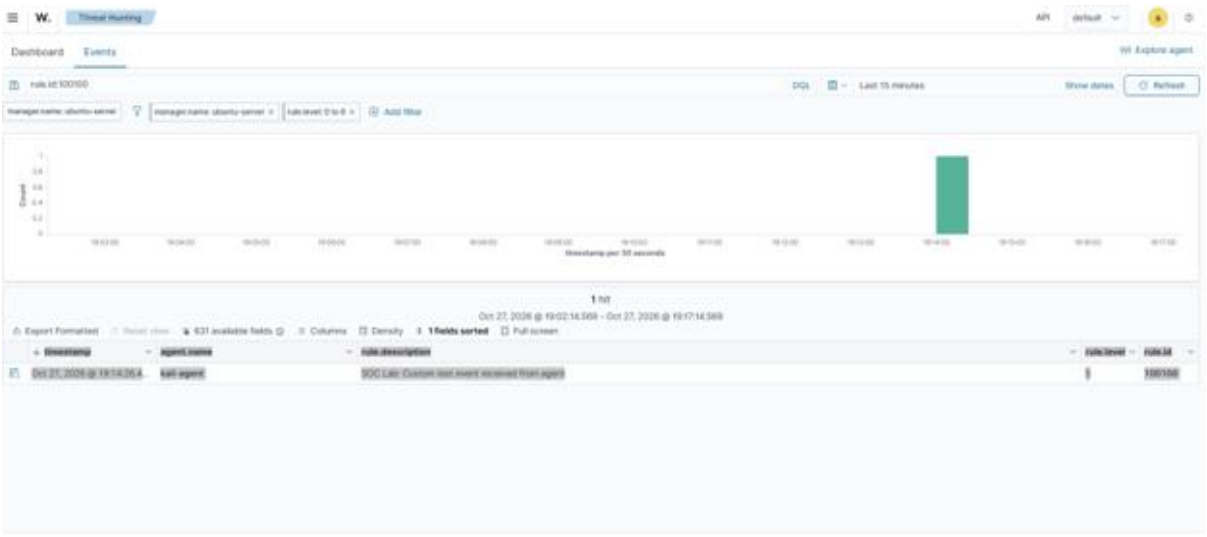
The following screenshots were provided for the project evidence. Keep the original files in the GitHub repository and ensure the filenames used in README.md match the uploaded files exactly.

Dashboard overview



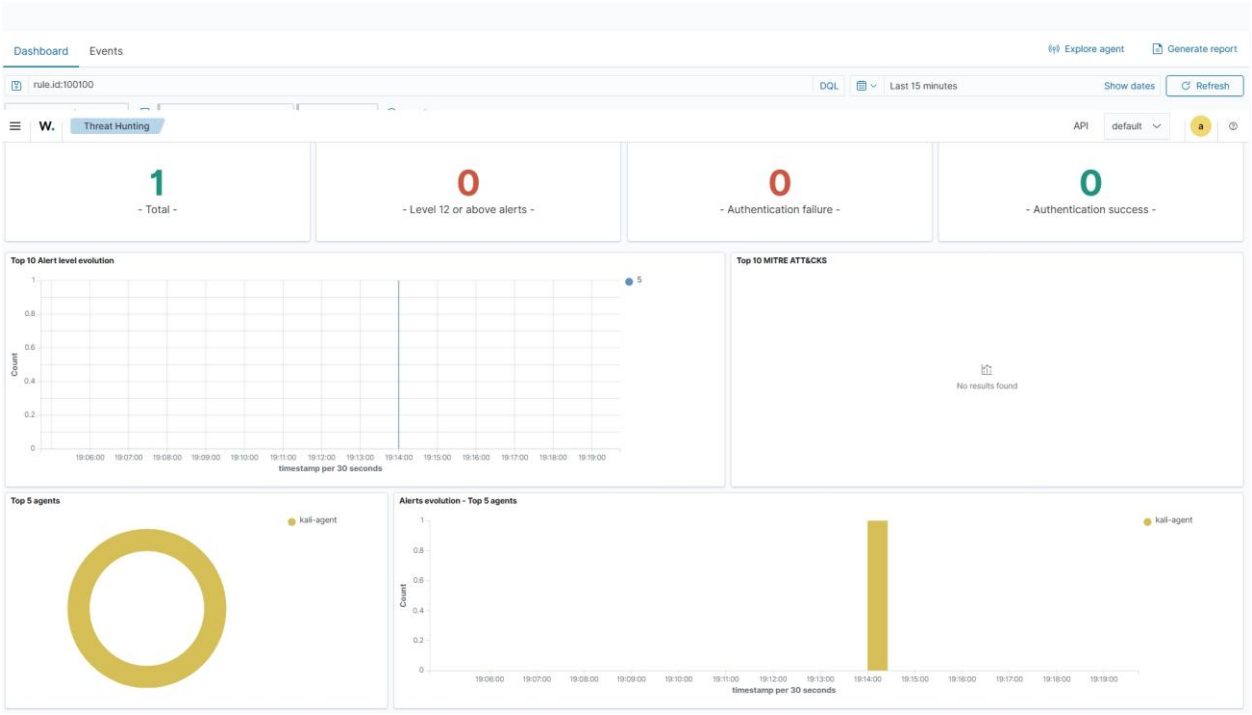
Screenshot_27-10-2026_192150_192.168.31.44.jpeg

Custom alert event in Threat Hunting → Events



Screenshot_27-10-2026_19182_192.168.31.44.jpeg

Threat Hunting dashboard filtered for kali-agent



Screenshot_27-10-2026_192036_192.168.31.44.jpeg

9. Troubleshooting Notes

Disk space

During setup, the Ubuntu root filesystem became almost full. Wazuh vulnerability-detection updater files under /var/ossec/queue/vd_updater/tmp/contents consumed several gigabytes. Journal

vacuuming and removal of a specifically reviewed large temporary updater file helped restore free space. Do not delete Wazuh queue or indexer data files indiscriminately.

```
df -h
sudo journalctl --vacuum-size=100M
sudo du -sh /var/ossec/queue/vd /var/ossec/queue/vd_updater
```

Service checks

```
sudo systemctl is-active wazuh-agent
sudo systemctl is-active wazuh-manager
sudo systemctl is-active wazuh-indexer
sudo systemctl is-active wazuh-dashboard
```

If any service is not active, collect the service status and recent logs before making changes. Avoid deleting /var/lib/wazuh-indexer or wiping indexes as a troubleshooting shortcut.

10. Results

- Wazuh dashboard was accessible.
- Kali agent kali-agent (ID 001) appeared as Active on the manager.
- Sudo activity from Kali appeared as Wazuh rule 5402 alerts.
- The custom marker event was detected by custom rule 100100 at level 5.
- Threat Hunting Events displayed the custom alert, and screenshots were saved as project evidence.

11. Conclusion

This project demonstrates the basic workflow of endpoint-based security monitoring using Wazuh: agent enrollment, event collection, rule matching, and dashboard verification. The custom alert test confirms that a locally defined detection rule can identify a specific test marker sent from the Kali endpoint.

For final submission, include this report, the original dashboard screenshots, and the GitHub repository URL. Before submitting, open each screenshot link on GitHub and verify that all images render.

Appendix A. Commands Quick Reference

```
# Ubuntu Server
lsb_release -a
df -h
ip addr
sudo systemctl is-active wazuh-indexer
sudo systemctl is-active wazuh-manager
sudo systemctl is-active wazuh-dashboard
sudo /var/ossec/bin/agent_control -lc
sudo grep -F 'Successful sudo to ROOT executed' /var/ossec/logs/alerts/alerts.json | tail -n
```

```
3
sudo grep -n -A 12 -B 5 '100100' /var/ossec/etc/rules/local_rules.xml
sudo grep -F 'SOC_LAB_TEST_EVENT' /var/ossec/logs/alerts/alerts.json | tail -n 3

# Kali Linux
sudo systemctl is-active wazuh-agent
logger "SOC_LAB_TEST_EVENT: Kali agent test successful"
```

Security note: never publish dashboard passwords, private keys, certificates, tokens, or other credentials in the repository or report.